

Kabadiwala Connect — Complete Recycler Journey

Purpose: This document explains the complete journey of the Recycler in Kabadiwala Connect: from creating the recycler profile and rates, through receiving a Dealer's lot, verifying the physical material and weight, confirming the handover, and finally maintaining the completed transaction/verified record.

The Recycler is the downstream verification and recycling-side actor. The Dealer prepares and sends a lot; the Recycler receives that lot, verifies what physically arrived, confirms the handover, and becomes the source of the downstream recycling-side record.

Core chain: Recycler Profile → Accepted Materials/Rates → Incoming Lot → QR/Manual Lookup → Lot Details → Physical Inspection → Verified Weight → Discrepancy Check → Confirm Handover → Transaction Completed → Verified Transaction / EPR-related Record → Recycler History.

1. UNDERSTANDING THE RECYCLER'S ROLE

The Recycler is the final operational participant in our main prototype flow. A Dealer aggregates material into a lot and chooses a Recycler. The Recycler then receives that lot physically and verifies that the digital information matches the material that arrived.

The Recycler dashboard therefore answers four questions: **What material is coming to me? Who sent it? What did they declare? What did I actually receive?**

The Recycler should not be treated as merely a person who clicks 'accept'. The verification step is what makes the handover traceable.

2. RECYCLER LOGIN

The Recycler opens the web dashboard and signs in. After authentication, the dashboard loads the recycler's profile, accepted materials, rates and incoming lots.

The dashboard should provide clear navigation such as: Profile/Rates, Incoming Lots, Verify Handover, Transaction History and Certificate/Records.

3. RECYCLER PROFILE

Before a Dealer can receive meaningful matching results, the Recycler needs to describe what they accept and what they offer.

The profile contains operational information such as recycler identity, accepted material categories, rate per category, pickup availability and service radius.

Fixed categories: PCB, CRT, LCD, Cable, Battery, Motor/Magnet and Mixed Plastic.

A Recycler should only have a rate associated with a category that it accepts. These values become inputs to the Dealer's recycler-matching feature.

4. SETTING ACCEPTED MATERIALS AND RATES

Example Recycler Profile:

PCB — Accepted — ■500/kg

Cable — Accepted — ■300/kg

LCD — Accepted — ■450/kg

Battery — Not accepted

The Recycler can also indicate pickup availability and the geographic/service radius. This lets the Dealer matching system rank suitable recyclers.

Important: the Recycler does not receive every possible lot. The Dealer's matching process should prefer recyclers whose accepted materials, rates, distance and pickup availability make sense.

5. DEALER CREATES A LOT

The Recycler does not create the lot. The Dealer does.

The Dealer selects available stock and creates a **single-category lot**, for example: LOT-123456, PCB, declared weight 30 kg.

The Dealer then selects the Recycler. The lot becomes associated with that Recycler and appears in the Recycler's Incoming Lots view.

6. INCOMING LOTS — RECYCLER HOME

The Recycler's main operational screen is the Incoming Lots list.

A possible row/card is:

LOT-123456 | PCB | 30 kg declared | Dealer XYZ | Pending Verification

The Recycler can open the lot to inspect its details. The list should make it obvious which lots are waiting, which have been verified and which are already completed.

7. HOW DOES THE LOT REACH THE RECYCLER?

The Dealer generates a QR code for the lot before physical handover.

QR rule: the QR contains only the Lot UUID. It does not carry the complete transaction payload.

The Recycler can identify the lot in two ways:

Option A: scan the QR.

Option B: manually enter the fallback lot/code if scanning is unavailable.

The backend then looks up the Lot UUID and returns the authoritative lot information.

8. QR SCAN / MANUAL LOOKUP

After successful lookup, the Recycler should see enough information to identify the expected shipment before confirming anything.

Example:

Lot ID: LOT-123456

Material: PCB

Dealer: ABC Dealer

Declared weight: 30 kg

Assigned Recycler: GreenCycle Recycling

Status: Pending Handover

The QR scan itself does **not** confirm the transaction. It only identifies the lot. Confirmation happens later, after physical verification.

9. PHYSICAL MATERIAL ARRIVES

Now the digital flow meets the physical world. The Recycler receives the scrap corresponding to the lot.

The Recycler should inspect the material and compare it with the digital lot: category, approximate quantity and physical condition/relevant information.

The dashboard should keep the expected/dealer-declared values visible while the Recycler enters what was actually verified.

10. VERIFIED WEIGHT

The Recycler physically weighs the material and enters the measured value into the dashboard.

Example:

Dealer declared: 30 kg

Recycler verified: 28 kg

These must remain two separate fields. The system must preserve the Dealer's original declaration instead of overwriting it with the Recycler's measurement.

This gives the transaction a clear audit trail: what was declared versus what was actually received/verified.

11. WEIGHT DISCREPANCY CALCULATION

The system compares the declared and verified weights.

Conceptually:

Discrepancy % = $|\text{Declared} - \text{Verified}| \div \text{Declared} \times 100$

Example:
Declared = 30 kg
Verified = 20 kg
Discrepancy = 33.3%

If the discrepancy is greater than **30%**, the dashboard shows a warning.

12. WHAT HAPPENS WHEN DISCREPANCY IS >30%?

The warning should be visible and understandable:

■ Weight discrepancy detected
Declared: 30 kg
Verified: 20 kg
Difference: 33.3%

The warning does not automatically reject the transaction. The Recycler can review the measurement and still continue to confirmation.

This is a deliberate business rule. The backend must enforce the same rule as the dashboard so that the UI and API do not disagree.

13. CONFIRM HANDOVER

Once the Recycler is satisfied with the material and verified weight, they select **Confirm Handover**.

The confirmation request goes to the backend. The backend checks that the lot exists, is associated with the correct workflow, has not already been confirmed and contains the required verification information.

On success, the handover becomes a confirmed transaction.

14. DUPLICATE CONFIRMATION

A mobile/web network can cause a user to retry a request. The system must prevent the same lot from being confirmed twice.

First confirmation → SUCCESS.

Second confirmation → REJECTED: Lot already confirmed.

This protection belongs on the backend. Disabling a button in the UI alone is not sufficient.

15. LOT STATUS AFTER CONFIRMATION

Once confirmation succeeds, the lot moves out of the pending handover state and becomes a completed/sold transaction according to the shared status model.

The Dealer should see the same completed status. The Recycler should also see the completed transaction in its history.

The status change is important because it prevents the same lot from continuing to appear as an available/pending shipment.

16. TRANSACTION HISTORY

The Recycler should be able to view completed transactions after confirmation.

Example:
LOT-123456 — PCB — Declared 30 kg — Verified 28 kg — Dealer ABC — Completed

History provides an operational record and makes it easier to find previous lots and their verified weights.

17. VERIFIED TRANSACTION / EPR-RELATED RECORD

After a confirmed handover, Kabadiwala Connect can generate a platform-side verified transaction/traceability record from the confirmed transaction data.

The record can contain: lot ID, transaction reference, Dealer, Recycler, material category, declared weight, verified weight, date/time, location if available and relevant transaction information.

Important: this platform-generated PDF should not be represented as an official government EPR certificate unless the project actually integrates with the authorized official EPR system. It is a verified transaction/traceability record that can support downstream records and reconciliation.

18. WHAT THE RECYCLER SEES — FULL SCREEN FLOW

Dashboard Home
↓
Incoming Lots
↓
Select LOT-123456
↓
Scan QR / Enter Code
↓
Lot Details
↓
Physical Material Received
↓
Enter Verified Weight
↓
Compare Declared vs Verified
↓
If >30%: Show Warning
↓
Confirm Handover
↓
Backend Confirms
↓
Lot Completed
↓
Transaction History + Verified Record/PDF

19. COMPLETE REALISTIC EXAMPLE

Step 1 — Recycler profile: GreenCycle accepts PCB and offers ■500/kg.

Step 2 — Dealer creates lot: Dealer has 30 kg PCB and creates LOT-123456.

Step 3 — Matching: GreenCycle is selected based on accepted material, rate, distance and pickup availability.

Step 4 — Incoming lot: LOT-123456 appears in GreenCycle's dashboard.

Step 5 — Handover: Dealer shows the QR and physically hands over the material.

Step 6 — Lookup: Recycler scans the QR; backend retrieves LOT-123456.

Step 7 — Verification: Recycler weighs the material and gets 28 kg.

Step 8 — Comparison: Declared 30 kg vs verified 28 kg. The values are both retained.

Step 9 — Confirmation: Recycler confirms the handover.

Step 10 — Completion: The lot becomes completed/sold, Dealer and Recycler histories update, and a platform verified transaction record/PDF can be generated.

20. WHAT HAPPENS IF SOMETHING GOES WRONG?

Situation	Expected behavior
QR cannot be scanned	Allow manual lot/code entry.
Lot UUID not found	Show clear not-found error; do not create a transaction.
Wrong recycler tries to process lot	Backend validates the lot/recycler association and rejects unauthorized confirmation.
Verified weight differs by >30%	Show warning; still allow confirmation.
Recycler presses Confirm twice	First succeeds; second is rejected as already confirmed.
Network fails during confirmation	Do not create a duplicate; retry safely using the same request identity/idempotency approach.

Situation	Expected behavior
Required verification data missing	Block confirmation and identify the missing field.
Lot already completed	Do not show it as a pending lot or allow another confirmation.

21. RECYCLER DASHBOARD MODULES

Module	Purpose
Login	Authenticate Recycler.
Profile/Rates	Declare accepted categories, rates, pickup availability and service area.
Incoming Lots	Show Dealer lots awaiting receipt/verification.
Lot Lookup	Find a lot using QR or manual code.
Lot Details	Show authoritative Dealer/lot information.
Weight Verification	Record the physical verified weight.
Discrepancy Check	Compare declared and verified weight and warn when >30%.
Confirm Handover	Complete the transaction after verification.
Transaction History	View completed lots.
Certificate/Records	View/export platform verified transaction record.

22. RECYCLER'S RESPONSIBILITY VS DEALER'S RESPONSIBILITY

Dealer: records what was purchased, maintains stock, creates the lot, chooses the recycler and initiates handover.

Recycler: receives the physical lot, looks it up, verifies the material/weight, records the verified weight and confirms the handover.

Backend: keeps the authoritative state, prevents invalid/duplicate confirmations and connects the two sides.

This separation is important. The Recycler should not edit the Dealer's original declared weight, and the Dealer should not be able to mark a Recycler handover as confirmed on the Recycler's behalf.

23. RECYCLER DATA FLOW

Recycler Profile/Rates → matching data → Dealer selects Recycler → Lot assigned → Incoming Lot → QR/Manual Lookup → Lot data → Physical Verification → Verified Weight → Discrepancy Rule → Confirm Handover → Transaction → History + Verified Record.

24. FINAL RULES THE RECYCLER DEVELOPER MUST NOT BREAK

- Use the same seven material categories as the Dealer and ML systems.
- QR contains only the Lot UUID.
- QR scanning identifies a lot; it does not itself confirm the transaction.
- Declared weight and verified weight are separate fields.
- >30% discrepancy produces a warning but does not automatically block confirmation.
- Already-confirmed lots cannot be confirmed again.
- The backend, not only the UI, enforces ownership/status/duplicate rules.
- Use the platform's verified transaction record as a traceability artifact; do not claim it is an official government EPR certificate without an actual authorized integration.
- Do not hard-code a fake successful transaction for the final demo; the lot should move through the real dashboard flow.

25. ONE-SENTENCE EXPLANATION FOR TEAMMATES

The Recycler dashboard receives a Dealer's lot, identifies it through its Lot UUID, lets the Recycler compare the expected shipment with the physical material, records the verified weight, warns about major discrepancies, confirms the handover once, and preserves the completed transaction as a traceable record.